

# Debian Server & Apache Lab Report

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## 1. Server Hardware Specifications (Virtual Machine Settings)

[img](#)

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## 2. Debian Login Screen

[img](#)

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## 3. IP Address of Debian Server Virtual Machine

[img](#)

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## 4. Working with the Firewall in Debian

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Debian commonly uses **UFW (Uncomplicated Firewall)**.

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### Check if the Firewall is Running

Command

```
sudo ufw status
```

Explanation

- **sudo** → Runs the command with administrator privileges.
- **ufw** → Firewall management tool.
- **status** → Displays whether the firewall is active or inactive.

Example Output

```
Status: active
```

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### Disable the Firewall

Command

```
sudo ufw disable
```

### Explanation

Disables the firewall completely.

### Example Output

```
Firewall stopped and disabled on system startup
```

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## Add Apache to the Firewall

### Command

```
sudo ufw allow 'Apache'
```

### Explanation

Allows HTTP web traffic through the firewall for Apache.

## Verify Firewall Rules

```
sudo ufw status
```

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# 5. Apache Commands and Administration

Apache service name in Debian:

```
apache2
```

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## Check if Apache is Running

### Command

```
sudo systemctl status apache2
```

## Explanation

- `systemctl` → Controls Linux services.
  - `status` → Displays whether Apache is running.
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## Stop Apache

### Command

```
sudo systemctl stop apache2
```

### Explanation

Stops the Apache web server service.

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## Restart Apache

### Command

```
sudo systemctl restart apache2
```

### Explanation

Restarts Apache after configuration change.

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## Test Apache Configuration

### Command

```
sudo apache2ctl configtest
```

### Explanation

Checks Apache configuration files for syntax errors.

### Example Output

```
Syntax OK
```

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# Check Installed Version of Apache

## Command

```
apache2 -v
```

## Explanation

Displays the installed Apache version.

## Example Output

```
Server version: Apache/2.4.x
```

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## 6. Common Apache Configuration Files

| File/Directory                | Purpose                                      |
|-------------------------------|----------------------------------------------|
| /etc/apache2/apache2.conf     | Main Apache configuration file               |
| /etc/apache2/sites-available/ | Stores available virtual host configurations |
| /etc/apache2/sites-enabled/   | Stores enabled websites                      |
| /etc/apache2/ports.conf       | Defines listening ports                      |
| /etc/apache2/mods-available/  | Available Apache modules                     |
| /etc/apache2/mods-enabled/    | Enabled Apache modules                       |

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## 7. Apache Log Locations

| Log File                    | Purpose                |
|-----------------------------|------------------------|
| /var/log/apache2/access.log | Records website visits |
| /var/log/apache2/error.log  | Records Apache errors  |

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## 8. Commands to Review Apache Logs

### View Entire Log File

#### Command

```
cat /var/log/apache2/access.log
```

### Explanation

Displays the complete log file.

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## View Log One Page at a Time

### Command

```
less /var/log/apache2/access.log
```

### Explanation

Lets you scroll through logs interactively.

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## View Last 10 Lines

### Command

```
tail /var/log/apache2/error.log
```

### Explanation

Shows the newest log entries.

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## Monitor Logs in Real Time

### Command

```
tail -f /var/log/apache2/access.log
```

### Explanation

Continuously displays new log entries as they are added.

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